Breaking Bored? SciMemeX: Towards Automatic Generation of Scientific Memes from Research Articles

Sandeep Kumar, Asif Ekbal
Department of Computer Science and Engineering, Indian Institute of Technology Patna, India
(sandeep_2121cs29,asif)@iitp.ac.in

Abstract

The rapid growth of scientific literature makes it increasingly difficult to communicate a paper's core novelty in a concise and accessible manner. While prior work has explored visual summaries such as graphical abstracts, emerging informal formats such as scientific memes remain largely unstudied from a computational perspective. Scientific memes rely on concise visual–textual juxtaposition and contrast to highlight how new work differs from prior approaches, making them a challenging test case for controlled abstraction, contrastive reasoning, and creative compression in generative models. We formalize scientific meme generation as a novel multimodal, contrastive generation task and propose *SciMemeX*, a modular multi-agent framework that extracts core innovations, selects contextually appropriate meme templates, and iteratively generates and refines captions through contrastive feedback-guided refinement. We evaluate generated memes along three complementary dimensions—*scientific fidelity*, *clarity*, and *engagement potential*—and conduct extensive automatic and human evaluations on 1.5k computer science papers. Our results show that SciMemeX consistently outperforms strong single-agent prompting and existing multi-agent baselines across all evaluation dimensions, suggesting that scientific meme generation provides a constrained setting for exploring controlled creative generation in scientific communication context. We will make the dataset and code public[1].

1 Introduction

Scientific progress relies not only on the discovery of new knowledge, but also on the effective communication of research contributions to audiences (Leshner, 2003; Marín-González et al., 2017). As the volume of scientific publications continues to grow, researchers face increasing difficulty in conveying the novelty and significance of their work in a concise and accessible manner (Warren, 2003). To address this challenge, the scientific community has developed a range of compressed and visual communication formats, including abstracts, short talks, posters, and graphical or visual abstracts. Prior studies show that such visual formats can substantially increase online visibility and engagement with scientific articles, particularly on social media platforms (Oska et al., 2020). In parallel, platforms such as Twitter (X) and LinkedIn have become important channels through which researchers contextualize and disseminate their works beyond traditional academic venues (Bik and Goldstein, 2013; Van Eperen and Marincola, 2011).

Within this evolving communication landscape, *memes* have emerged as a widely used form of digital expression characterized by concise visual–textual juxtaposition, contrast, and often humor. Prior work in education and communication studies shows that meme-based artifacts can support learning, recall, and engagement, and are increasingly viewed as a stable form of online discourse and digital literacy (Riser et al., 2020; Shibilski, 2021; Myers, 2022). Recent practitioner accounts further document the use of science memes in conference presentations, teaching, blogs, and social-media outreach, suggesting that memes are increasingly being explored as informal science communication tools (Joseph, 2023; Tran, 2024). These findings suggest that memes may plausibly function as lightweight, contrastive explanatory artifacts in informal or semi-formal scientific communication settings. Despite these emerging uses, *scientific memes* remain largely unstudied from a computational perspective. There is currently no systematic framework that formalizes what constitutes a scientific meme, how such artifacts encode a paper's conceptual novelty, or whether generative models can produce them while preserving scientific fidelity.

[1] https://anonymous.4open.science/r/SciMeme-5FE1/

Rather than assuming memes to be a universally validated dissemination medium, we treat them as an *emerging artifact class* whose structure and generative feasibility warrant systematic investigation.

Definition: Scientific Meme

In this paper, we define *scientific memes* as concise visual–textual artifacts designed to communicate a paper's key novelty through contrastive framing. Unlike typical internet memes, which primarily emphasize humor and virality, scientific memes prioritize scientific fidelity, clarity, and memorability. Their goal is not entertainment but compact communication of scientific novelty, enabling readers to quickly grasp how a new work differs from prior approaches.

Unlike graphical abstracts, which aim for broad coverage of a study's content, scientific memes focus on contrastive compression—isolating and juxtaposing what is new against what came before. Importantly, we do not treat scientific memes as general-purpose internet humor artifacts. Rather, we target domain-literate scientific communication, where humor is optional and contrastive novelty representation remains the core structural requirement. Following prior work that treats internet memes as template-based, genre-like artifacts shaped by recognizable conventions (Shifman, 2013; Wiggins and Bowers, 2015), we focus on a contrastive "before vs. after" / "prior vs. proposed" structure, which highlights how a new paper differs from prior approaches. This format aligns with well-established cognitive principles, including dual coding theory (Paivio, 1990), contrastive framing effects (Kahneman and Tversky, 1984), and the role of surprise in memorability (Górka, 2014). By juxtaposing prior limitations with new contributions in a compact visual–textual form, such memes offer a concise representation of scientific novelty well suited to fast-paced digital environments. Beyond its application context, scientific meme generation poses challenges that are independently relevant to natural language processing and multimodal generation. Producing a faithful scientific meme requires models to identify a paper's central contribution, reason contrastively with respect to prior work, compress this distinction into a highly constrained visual–textual format, and balance creativity with factual correctness. These requirements go beyond standard summarization or captioning tasks, instead demanding controlled abstraction, contrastive reasoning, and creative compression—capabilities that remain challenging for current large language models. As such, scientific meme generation serves as a compact testbed for studying these fundamental capabilities, regardless of the eventual adoption of memes as a dissemination medium.

In this paper, given a scientific paper, our goal is to automatically generate a meme that accurately and clearly communicates the paper's key novelty relative to prior work while remaining faithful to the underlying scientific content. To this end, we introduce **SciMemeX**, an agentic framework for generating scientific memes directly from research papers. SciMemeX integrates multiple specialized agents that collaboratively extract core contributions, select contextually appropriate meme templates, and generate contrastive captions through an iterative refinement process.

Our contributions are *threefold*: **(i)** we formalize scientific meme generation as a novel multimodal, contrastive generation task that probes controlled abstraction and creative compression in generative models; **(ii)** we propose **SciMemeX**, an agentic framework that decomposes the task into contribution extraction, template selection, and iterative contrastive feedback-guided meme generation; and **(iii)** through comprehensive automatic and human evaluations, we show that our approach consistently outperforms single-agent prompting and existing multi-agent baselines in terms of scientific fidelity, clarity, and engagement.

2 Related Work

Prior work on scientific communication has explored a range of textual (Hermann et al., 2015; Sefid and Giles, 2022; Sharma et al., 2024; Dong et al., 2021) and multimodal abstractions, including summaries (Tan et al., 2025; Kumar et al., 2024a), graphical abstracts (Kawada et al., 2025), posters (Wang et al., 2024b; Tanaka et al., 2024; Landis and Duscher, 2022) , and slide-style presentations (Sefid et al., 2021), aimed at conveying a paper's core contributions in a concise and accessible form. More recently, P2P (Sun et al., 2025) used coordinated LLM agents to generate poster-style visual summaries of papers. However, such methods organize paper content rather than foreground novelty in brief, engaging formats, motivating more concise communication artifacts. Prior work on automatic meme generation has largely targeted general-purpose social media content, emphasizing humor and virality over factual or technical accuracy (Vyalla and Udandarao, 2020; Wang and Lee, 2024a).

While recent vision language models enable more context aware meme generation, they primarily target general purpose social media content and do not account for scientific fidelity or the need to distinguish new contributions from prior work. To the best of our knowledge, no prior work addresses scientific meme generation as a dedicated scientific communication task. We hope this work inspires future research at the intersection of AI-driven creativity and accessible science dissemination.

3 Methodology

3.1 Problem Definition

We define scientific meme generation as the task of producing a meme that communicates a paper's key novelty relative to prior work. Let a paper be p (abstract, sections), a template set $\mathcal{M}$, and a caption space $\mathcal{C}$. A meme is $m = (t, c)$ with $t \in \mathcal{M}$, $c \in \mathcal{C}$, contrasting prior work and new contributions. A team of agents $\mathcal{A} = \{$ Concisio, Template_Selector, Generator, Critics$\}$ induces a policy $\pi_{\mathcal{A}}(m \mid p)$ realized through an exploration and exploitation contrastive feedback loop. Rather than formulating a directly solved optimization problem, we define a conceptual objective that guides the iterative generation and refinement process. In practice, candidate memes are assessed along three desiderata, namely scientific fidelity, clarity and interpretability, and engagement. These criteria are used by the critique agents to compare and refine candidates during the search process. The final output corresponds to the best candidate identified through this feedback driven procedure.

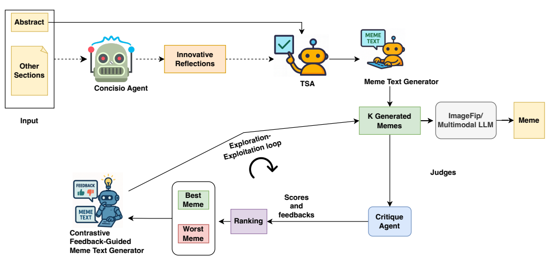


Figure 1: Agentic framework for meme generation. The process consists of an initial **exploration** phase (diverse template selection and generation) followed by iterative **exploitation** through contrastive refinement, guided by critique feedback.

3.2 Framework Overview

Grounded in the dual perspective of human creative cognition and computational exploration–exploitation, our methodology operationalizes this cycle within an agentic computational framework. SciMemeX instantiates alternating phases of divergent ideation and convergent critique through a coordinated ensemble of agents that specialize in generation, selection, and evaluation. Each agent embodies a cognitive role—idea condensation, template prioritization, caption generation, or multi-axis critique—collectively forming an adaptive reasoning loop. This design unites psychological models of creativity (Guilford, 1967; Torrance, 1988; Sawyer, 2012) with algorithmic principles of adaptive search (Sutton and Barto, 2018; Kaelbling et al., 1996), framing scientific meme generation as both a cognitive simulation and an optimization process (Dawkins, 1976; Blackmore, 1999). The workflow of the proposed SciMemeX framework is illustrated in Figure 1, with the detailed algorithm provided in Appendix 1. The input consists of a paper's abstract and, optionally, its other sections. If only the abstract is available, the *Concisio Agent* is bypassed; otherwise, the *Concisio Agent* summarizes the paper and extracts key innovative reflections, which are then passed to the *Template Selector Agent (TSA)*. The TSA identifies suitable meme templates and forwards them to the *Meme Text Generator*, which produces k captioned meme candidates. Each candidate is evaluated independently by three *Critique Agents*—*Fidelity Critique*, *Clarity Critique*, and *Engagement Critique*—whose scores are used to rank the memes. The best- and worst-performing memes are then provided to the *Contrastive Feedback-Guided Generator*, which refines new captions by integrating both positive and negative feedback. This exploration–exploitation loop repeats until convergence or a predefined iteration limit. Finally, the selected template–caption pair is rendered into the final meme image.

In the following sections, we describe the principal components of **SciMemeX**, including the meme-generation agents, their role specifications, and the inter-agent communication strategy. The proposed multi-agent system operates within an exploration–exploitation loop: agents first extract a contrastive summary from the input paper p, select suitable templates, generate captioned meme candidates, and iteratively refine them based on critique feedback until convergence or a predefined computational budget is reached. These meme-generation agents—*Concisio*, *Template Selector*, *Generator*, and *Critique*—form the core of our framework, collectively enabling adaptive reasoning and creative synthesis across iterative refinement cycles. SCIMEMEX is a training-free inference-time framework that coordinates specialized LLM agents via contrastive comparison and feedback-guided refinement.

3.3 Concisio Agent

The Concisio Agent integrates comparative analysis and concise synthesis into a reasoning module. Given a research paper $\mathcal{R}$, the agent first performs a comparison between prior and current methodologies[2]. It identifies a set of prior approaches $\mathcal{M}_{\text{old}} = \{M_{p_1}, \ldots, M_{p_n}\}$ and extracts the main contribution(s)[3] of the current work as M_{new}. For each $M_{p_i} \in \mathcal{M}_{\text{old}}$, the agent identifies conceptual and technical distinctions Δ_i between M_{new} and M_{p_i}, capturing differences in mechanism, objective, or theoretical framing. Based on this comparison, the agent produces a compact summary S^* from $\mathcal{M}_{\text{old}}$, M_{new}, and the set of distinctions $\{\Delta_i\}$. The summary S^* captures (i) how the problem was addressed previously, (ii) what the current paper does differently, and (iii) why this difference is meaningful. We operationalize a paper's *key contribution* as contrastive novelty relative to prior work: what prior approaches did, what the current work changes, and why the difference matters. The output is organized into three interpretable segments, *Prior Work (Old Ideas)*, *Proposed Approach (New Idea)*, and *Core Differences*, and expressed as a concise narrative of approximately 100 words. This agent provides a clear and faithful foundation for downstream modules such as the Template Selector and Meme Generator.

3.4 Template Selector Agent

The Template Selector Agent (TSA) is a prompt based LLM module that uses the innovation summary S^*, consisting of the abstract and extracted reflections, to identify suitable meme templates from the set $\mathcal{M}$. We define $\mathcal{M}$ as a fixed, externally curated pool of 250 meme templates collected from publicly available meme-template repositories and websites, including ImgFlip[4] and other open meme-template catalogs, after filtering. We retain templates containing at least two editable text placeholders and a clear visual structure suitable for contrastive framing, while excluding single-caption, visually unclear, or non-contrastive "prior work vs. proposed work" formats. Given S^*, the agent selects a top k subset $T_p \subseteq \mathcal{M}$ that is appropriate for highlighting the contrast between prior work and the new contribution.

3.5 Meme Generation and Refinement

As illustrated in Figure 1, SCIMEMEX conducts meme generation through two complementary stages—initial exploration and feedback-guided refinement—forming an adaptive exploration–exploitation loop grounded in both human creative cognition and algorithmic search.

Initial Exploration. Conditioned on the innovation summary S^* and the selected template set T_p, the Meme Text Generator produces k captioned candidates per template. This phase emphasizes diversity and breadth, mirroring human *brainstorming*: numerous unconstrained captions are generated to cover varied humorous framings and contrasts between prior and new work. The resulting candidates form the initial pool for evaluation.

Feedback-Guided Refinement. Each candidate is scored by the critique agent, which also provides textual feedback. The highest- and lowest-ranked memes (m^+, m^-) serve as contrastive anchors for the Feedback-Guided Meme Generator, which synthesizes a new batch of k refined captions that retain the strengths of m^+ while avoiding the weaknesses of m^-. Although this stage primarily exploits prior feedback, it preserves controlled exploration via stochastic prompting and stylistic variation. The loop iterates until convergence or a budget cap, returning the highest-scoring candidate $m^\star$.

Finally, the selected meme template and generated caption are rendered into the final meme image. We use ImgFlip as the default renderer whenever the selected template is available in its public catalog; this accounts for 86% of the generated memes in our evaluation. For the remaining 14%, where the selected template is not available in ImgFlip or requires a custom multi-panel composition, we use GPT-4o as the fallback renderer. We selected this multimodal renderer once at implementation time based on pilot checks for prompt following, template adherence, and readable text placement;

[2]Concisio relies only on information explicitly provided in the input paper rather than external model knowledge, reducing the risk of hallucinated prior-work comparisons.

[3]Many papers contain multiple complementary contributions. In such cases, Concisio extracts multiple contrastive conceptual and technical distinctions relative to prior work and synthesizes the most salient innovations into a unified compact summary.

[4]https://imgflip.com/

the renderer is not chosen on a per-paper basis. Importantly, SCIMEMEX outputs a template–caption pair, so the agentic generation pipeline is rendering-agnostic. During iterative refinement, we intentionally keep the meme template fixed and refine only the caption. This design ensures that critique feedback remains grounded in a stable visual framing; changing templates mid-loop would invalidate the critics' judgments and led to unstable or incoherent revisions in pilot experiments.

4 Evaluation

A central challenge in this work is to ensure that the proposed evaluation framework is both theoretically sound and practically reliable. To this end, we designed a three-phase evaluation pipeline that progressively integrates expert insights, LLM-as-judge validation, and final human benchmarking. This pipeline was constructed to address common criticisms of evaluation in generative tasks, namely: (i) overreliance on ad hoc human judgments, (ii) the limited generalizability of traditional metrics (e.g., BLEU/ROUGE), and (iii) the lack of systematic validation for LLM-based scoring.

4.1 Phase I: Expert-Driven Metric Design

We began with a structured pilot study involving twelve senior experts, including mid-career faculty, postdoctoral researchers, and PhD candidates (5–10 years of experience) with more than 5+ ACL/ICLR or related publications. Experts evaluated memes generated for a representative set of 80 papers drawn from their own publication history and from comparable peers. Rather than limiting themselves to one-shot generations, each expert engaged in an interactive human-in-the-loop process: they prompted the LLM of their choice (CHATGPT, GEMINI, or CLAUDE) (Achiam et al., 2023; Anthropic, 2023; DeepMind, 2023), inspected its outputs, and iteratively refined the memes with targeted feedback until they were satisfied that the result captured the paper's core contribution. Across four workshop rounds, experts not only judged the resulting memes but also critiqued and revised them, using the iterative process itself as a way to stress-test evaluation criteria. In doing so, they revealed both the strengths and limitations of LLMs when steered by human expertise, and distilled a set of evaluation dimensions that remained robust even under this adversarial, feedback-driven setting.

We observed consistent failures of lexical overlap metrics (e.g., BLEU, ROUGE) (Papineni et al., 2002; Lin, 2004) and vision–language similarity scores (Radford et al., 2021a) to capture the desiderata of our task. Through structured annotation sessions and consensus-building (Artstein and Poesio, 2008a; Krippendorff, 2018), experts distilled three dimensions as essential and non-redundant: (i) **Scientific Fidelity**, (ii) **Clarity & Interpretability**, and (iii) **Engagement Potential**. Importantly, the design process reached saturation: after four workshop rounds, no additional dimensions were proposed. Scientific Fidelity and Engagement Potential were each operationalized on a **1–5 Likert scale**, allowing fine-grained differentiation. In contrast, Clarity & Interpretability employed a **3-point scale**, as pilot testing showed that annotators could not reliably sustain more granular judgments in this dimension without over-interpretation. We discuss the details of the evaluation metrics and their definitions in the appendix C.

4.2 Phase II: LLM-as-Judge Validation

Having formalized the rubrics, we implemented them as structured prompts and evaluated their reproducibility under an LLM-as-judge paradigm (Cai et al., 2025) using GPT-4.1 (Achiam et al., 2023) on a new sample of 300 memes. Six independent domain experts (senior PhD students and early-career researchers, not involved in Phase I) annotated each meme on all three dimensions, with three raters per item.

Inter-annotation agreement was measured using standard reliability metrics (Artstein and Poesio, 2008a; Krippendorff, 2018), yielding substantial human–human consistency ($\tau = 0.69$, $\kappa = 0.66$, $r = 0.72$, $\alpha = 0.68$). GPT-4.1, prompted with the same rubric under controlled settings, was then applied to the identical dataset. Its ratings showed high concordance with human judgments ($\tau = 0.71$, $\kappa = 0.68$, $r = 0.74$). These values fall within or above commonly accepted thresholds for strong inter-annotator reliability in computational linguistics (Artstein and Poesio, 2008b; Krippendorff, 2018). Importantly, the LLM-as-judge framework achieved higher internal consistency than additional human raters with limited domain expertise, reinforcing its validity as a scalable proxy. Prompt templates, Annotator Guidelines, Annotation Training, Annotation Steps are discussed in detail Appendix E.

Model	Method	Fidelity	Clarity	Engagement	Total
Single-Agent Baselines (CoT Prompting)					
Mistral-7B (I)	CoT	2.84	1.70	3.18	7.72
Llama-3.1-8B (I)	CoT	3.02	1.77	3.30	8.09
Llama-3.1-70B (I)	CoT	3.12	1.82	3.47	8.41
Claude-3.5 Sonnet	CoT	3.25	1.86	3.60	8.71
Gemini-1.5 Pro	CoT	3.34	1.89	3.72	8.95
GPT-4o	CoT	3.48	1.93	3.85	9.26
Template-Based Meme Systems					
Memeify	Template-Captioning	2.21	1.41	2.96	6.58
MemeCraft	Template-Captioning	2.48	1.55	3.12	7.15
Multi-Agent Frameworks					
Llama-3.1-70B (I)	MAD	3.17	1.85	3.56	8.58
	Self-Reflect	3.32	1.92	3.68	8.92
	ChatEval	3.45	1.95	3.77	9.17
	Self-Refine	3.56	1.97	3.74	9.27
	SciMemeX (Ours)	**4.31**	**2.09**	**3.90**	**10.30**
Claude-3.5 Sonnet	MAD	3.26	1.87	3.67	8.80
	Self-Reflect	3.41	1.94	3.75	9.10
	ChatEval	3.53	1.97	3.83	9.33
	Self-Refine	3.67	2.00	3.82	9.49
	SciMemeX (Ours)	**4.44**	**2.12**	**3.96**	**10.52**
Gemini-1.5 Pro	MAD	3.34	1.90	3.72	8.96
	Self-Reflect	3.54	1.98	3.78	9.30
	ChatEval	3.67	2.00	3.83	9.50
	Self-Refine	3.79	2.03	3.89	9.71
	SciMemeX (Ours)	**4.52**	**2.14**	**4.00**	**10.66**
GPT-4o	MAD	3.52	1.94	3.85	9.31
	Self-Reflect	3.70	2.00	3.88	9.58
	ChatEval	3.83	2.03	4.04	9.90
	Self-Refine	3.94	2.04	4.02	10.00
	SciMemeX (Ours)	**4.66**	**2.16**	**4.12**	**10.94**

Table 1: Automatic evaluation results across six open- and closed-source models released by July 2025. We report mean scores for Fidelity, Clarity, Engagement, and total composite scores for both single-agent (CoT) and multi-agent frameworks. "I" denotes instruction-tuned variants (Instruct models).

4.3 Phase III: Final Human Evaluation

The final phase benchmarked our framework against three closely related baselines (MAD, ChatEval, and Self-Refine) through a targeted human evaluation. We sampled 250 representative papers spanning diverse subfields (language modeling, machine translation, multimodal reasoning, and interpretability). To ensure consistency, we employed the same expert annotators from Phase II, who were already calibrated on the rubric but remained blind to system identity. All memes were rated on Fidelity (1–5), Clarity & Interpretability (1–3), and Engagement Potential (1–5), with human–human reliability monitored using Kendall's τ, Cohen's κ, and Krippendorff's α, and raw agreement reported without adjudication. Meme order was randomized within each paper to prevent bias, enabling a fair, head-to-head comparison of our framework against the baselines. We discuss the annotator's pay in Appendix E.1. This also serves as an additional safeguard against potential bias in LLM-based evaluation, as annotators were blind to system identity.

5 Experiments

We evaluate SciMemeX on 1.5k computer science papers randomly sampled from arXiv AI/NLP/ML categories from 2022–2023, using a 30/70 validation–test split. Papers were sampled uniformly from these categories, and we excluded only papers whose PDFs could not be parsed into usable title, abstract, and main-body text by GROBID (Developers, 2008). Automated evaluation is conducted on the held-out test split using the LLM-as-judge rubric validated in Section 4.2. We compare against single-agent prompting, template-based meme systems, and multi-agent baselines, including CoT, Memeify, MemeCraft, MAD, Self-Reflect, ChatEval, and Self-Refine. All baselines are evaluated with the same task inputs and the same underlying backbone model where applicable; full implementation and baseline details are provided in Appendices A and B.

5.1 Results

5.1.1 Automatic Evaluation

We report the results of SciMemeX and baselines using the automated metrics (discussed in 4.2) in

Table 1. We find that SCIMEMEX achieves the best scores across all metrics. The largest and most stable improvements appear in *Scientific Fidelity* (+0.72–+0.77 over the strongest baseline), confirming that our feedback-driven generation better preserves the underlying claims of the paper. *Clarity* also improves consistently (+0.11–+0.12), indicating that structured refinement enhances linguistic precision and interpretability. *Engagement* shows smaller, yet positive gains (+0.10–+0.16), suggesting that faithfulness and readability are not achieved at the cost of appeal. Template-based meme systems perform substantially worse across all metrics, reflecting the limitations of fixed template–captioning for scientific content. These improvements are consistent across model capacities—from Llama-70B to GPT-4o—demonstrating that SCIMEMEX generalizes beyond a specific architecture. In contrast to CHATEVAL and SELF-REFINE, which perform unguided multi-round refinements, our approach employs a structured exploration–exploitation strategy that explicitly contrasts the best and worst generations in each round. This contrastive critique–feedback loop enables targeted improvement across iterations rather than generic self-revision, allowing the model to progressively converge toward high-fidelity, interpretable, and engaging memes.

5.1.2 Human Evaluation

We now report the final human evaluation comparing SCIMEMEX against three strong baselines. Full details of the protocol are provided in Section 4.3. Table 2 summarizes mean human scores[5] with 95% confidence intervals, estimated via paired bootstrap resampling (1,000 draws over paper-level items). Statistical significance was assessed using paired bootstrap with Holm correction across metrics. SCIMEMEX consistently outperforms all baselines on all three dimensions. Improvements are significant for *Fidelity* ($p<.01$) and *Clarity & Interpretability* ($p<.05$), while *Engagement Potential* also shows a significant gain ($p<.05$), indicating that higher factual precision and clarity are achieved without compromising creative appeal. After normalization, SCIMEMEX achieves the highest composite score (2.43), reflecting a notable improvement of +0.43 over the strongest baseline. To verify that agreement was not driven by midpoint clustering, we report Phase III human score distributions in Appendix G.

[5]From the 250 papers sampled to construct a balanced pool, we selected a 100-paper subset.

5.2 Analysis

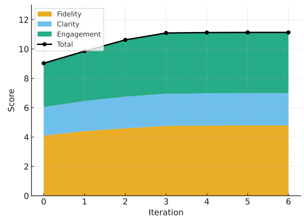


Figure 2: Stacked plot of scores across iterations. The black line shows the total score, while the stacked areas indicate the component contributions from Fidelity, Clarity, and Engagement.

5.2.1 Effect of Iterative Refinement

Figure 2 shows the progression of scores across successive critique–feedback iterations. We observe consistent improvements across all three evaluation dimensions during the early stages, with gains stabilizing after approximately the third iteration. *Scientific Fidelity* rises from 3.70 at iteration 0 to 4.76 by iteration 3 (+28.6%), reflecting improved preservation of core scientific claims through targeted refinement. *Clarity & Interpretability* improves from 2.00 to 2.19 (+9.5%), while *Engagement Potential* increases from 3.88 to 4.18 (+7.7%), showing that factual precision does not come at the cost of communicative appeal. Overall, the composite score improves from 9.58 to 11.13 (+16.1%), indicating that most benefits of iterative feedback occur within the first few rounds. This does not represent a hard architectural limit: the framework does not impose a fixed iteration cap, and further gains may arise from expanding the template space or critique signals. We discuss a refinement case study in Appendix M.

5.2.2 Template Diversity Analysis

Automatic meme generation systems often suffer from *template collapse*, repeatedly relying on a small number of formats. We therefore analyze template diversity on the Phase III evaluation set using Unique Template Count (UTC), template entropy (Shannon, 1948), and Top-1 template frequency (Holtzman et al., 2020). As shown in Table 3, SCIMEMEX exhibits higher diversity than single- and multi-agent baselines, with higher entropy and lower template dominance. This improvement arises from a single, contrast-aware template selection step; removing the Template Selector

System	Fidelity (1–5)	Clarity (1–3)	Engagement (1–5)	Norm. Total (0–3)
MAD	3.48 ± 0.11	1.92 ± 0.07	3.83 ± 0.12	1.78
ChatEval	3.84 ± 0.10	2.03 ± 0.06	4.04 ± 0.11	1.97
Self-Refine	3.97 ± 0.10	2.05 ± 0.06	4.05 ± 0.11	2.00
SciMemeX (Ours)	$\mathbf{4.68}\ ^{\star\star}_{\pm 0.09}$	$\mathbf{2.16}\ ^{\star}_{\pm 0.05}$	$\mathbf{4.21}\ ^{\star}_{\pm 0.09}$	$\mathbf{2.43}\ ^{\star\star}$
Δ vs. best baseline	+0.71	+0.11	+0.16	+0.43

Table 2: Phase III human evaluation on 100 papers (3 raters/item). Values are means with 95% confidence intervals (paired bootstrap, 1,000 resamples). $\star\ p<.05$, $\star\star\ p<.01$ (Holm-corrected). Normalized Total rescales each metric to [0,1] before summation: $\hat{F} = (F-1)/4$, $\hat{C} = (C-1)/2$, $\hat{E} = (E-1)/4$. Engagement improvements were statistically significant at $p<.05$.

System	UTC ↑	Entropy ↑	Top-1 (%) ↓
CoT (GPT-4o)	11	1.88	40.2
Memeify	9	1.74	45.1
Self-Refine	12	2.01	36.4
SciMemeX (Ours)	**17**	**2.39**	**23.6**
w/o TSA	13	2.05	33.8

Table 3: Template diversity analysis on the Phase III evaluation set. UTC denotes the number of unique meme templates used. Entropy measures the uniformity of the template distribution. Top-1 indicates the proportion of memes using the most frequent template.

Agent (w/o TSA) leads to reduced diversity. We discuss this analysis in detail in Appendix K.

Variant	Fid.	Cla.	Eng.	Total	Δ
Full	**4.66**	**2.16**	**4.12**	**10.94**	–
w/o Contrast.	4.55	2.09	3.95	10.59	-0.35
w/o Feedback	4.25	1.92	3.72	9.89	-1.05^{**}
w/o TSA	4.40	1.97	3.81	10.18	-0.76^{*}
w/o Concisio	4.50	2.02	3.98	10.50	-0.44
Abstract-only	4.32	1.95	3.88	10.15	-0.79^{*}

Table 4: Ablations for SciMemeX. Fid./Cla./Eng. denote Fidelity, Clarity, and Engagement. Δ is relative to the full system. Significance vs. full: $^{*}p<.05$, $^{**}p<.01$ (paired bootstrap).

5.2.3 Ablation Study

Table 4 summarizes the effect of removing key components from SciMemeX. Each ablation is evaluated independently against the full system. The steepest performance decline occurs when the iterative feedback loop is removed (*w/o Feedback*, -1.05), reaffirming that multi-round critique–revise interaction is central to the framework's effectiveness. Using only the abstract instead of the full paper context results in the second-largest reduction (-0.79), highlighting the importance of broader document context for accurate contribution extraction and meme generation. Removing the template selection agent (*w/o TSA*) also degrades performance (-0.76); in this setting, the informed top-k template selection is replaced with uniform random sampling of $k = 5$ templates from the same template pool $\mathcal{M}$, while keeping all other components unchanged. Smaller but consistent declines are observed when removing the Concisio agent (*w/o Concisio*, -0.44) or contrastive conditioning (*w/o Contrast.*, -0.35), indicating that both components provide complementary improvements.

Beyond the ablation evidence, we directly evaluated Concisio's contribution extraction on 40 randomly sampled papers. Two domain experts independently identified each paper's key contributions from the original manuscript, compared them with Concisio's generated summaries, and rated them on 1–5 Likert scales for Coverage (whether the main contributions were captured) and Correctness (whether unsupported or inaccurate claims were introduced). Concisio achieved average scores of 4.75 and 5.0 for Coverage and Correctness, respectively, indicating that its summaries preserve core contributions while avoiding hallucinated content. Overall, the gains of SciMemeX arise from iterative feedback, broader document context, informed template selection, and structured contrastive guidance. We provide error analysis of common failure cases in Appendix J.

6 Conclusion and Future Work

We introduce scientific meme generation, a task that translates research ideas into accessible visual–textual narratives. We propose SciMemeX, a modular agentic framework combining insight extraction, template selection, and feedback-guided refinement. Our three-phase evaluation pipeline measures Scientific Fidelity, Clarity, and Engagement through expert and LLM-based assessment. Experiments on 1.5k computer science papers show that SciMemeX significantly outperforms strong single- and multi-agent baselines, validating the benefits of contrastive feedback and structured coordination. Future work will extend SciMemeX across domains and support HCI-driven, feedback-adaptive human–AI co-creativity with interactive editing and constraint-guided control.

Limitations

We discuss the limitations of this work below:

- **Domain Scope**: Our experiments primarily focus on computer science papers (NLP/ML/AI). While this provides a controlled evaluation setting, extending SCIMEMEX to other disciplines such as biology or the social sciences may require domain-specific adaptation, which we leave for future work.

- **Challenges with Highly Technical Content**: Papers with dense mathematical formulations or abstract theories are often harder to translate into clear and humorous narratives. Such memes may assume expert-level familiarity with technical terminology. Incorporating audience modeling or adaptive simplification mechanisms could further democratize scientific communication[6].

- **Subjectivity of Humor and Engagement**: Humor, creativity, and engagement naturally vary across audiences and cultures. Although our three-phase evaluation pipeline mitigates this through expert calibration and LLM-as-judge validation, broader community-level studies could further enhance robustness and generalizability.

- **Concisio Validation Scope:** Although Concisio achieved high expert ratings in our 40-paper validation, this study used a small controlled sample and may reflect annotator calibration or reference-selection effects. Broader blinded validation across domains remains future work.

- **Computational Overhead**: Although our multi-agent framework substantially improves performance over single-agent baselines, implementing such systems inherently incurs higher computational and financial costs. Similar to other LLM-as-a-judge approaches, the processing efficiency and overall expense depend on the choice of underlying models. While our structured feedback and contrastive refinement mechanisms mitigate redundancy, achieving full cost efficiency and standardization across different LLM settings remains an open challenge.

- **Dynamic Template Adaptation:** An interesting extension is to adapt meme templates dynamically based on critic feedback. While promising, this would require re-aligning critic objectives with changing visual structures, as critique scores are conditioned on a fixed visual framing. Preliminary exploration suggested that switching templates mid-refinement can destabilize feedback and hinder convergence. We therefore leave dynamic template adaptation to future work.

- **Training-Free Design:** SCIMEMEX does not train task-specific model parameters and does not rely on supervised scientific-meme training data. While this makes the framework reproducible and applicable to a new low-resource task, future work could learn template-selection policies, reward models, or critique modules as larger datasets become available.

[6]While our current framework targets domain-literate audiences (as noted in Section 1), incorporating adaptive simplification could help bridge the gap to the general lay public.

Ethics

We explicitly designed our system to avoid generating offensive or disparaging content by incorporating safety critics and enforcing a hard rule whereby such outputs automatically receive the lowest engagement score. Nevertheless, humor is inherently subjective, and unintended biases may still emerge. We compensated annotators fairly in line with academic standards, and the study was reviewed under institutional policies and deemed exempt from IRB approval as it posed minimal risk and involved professional adult participants. We emphasize that SCIMEMEX is intended as a tool for broadening science communication and not for trivializing research or spreading misinformation. We used ChatGPT only for language polishing and proofreading the paper.

Copyright and Fair Use of Meme Imagery: The illustrative memes shown in this paper (Figures 1, 4–7) are based on widely circulated meme templates derived from copyrighted media. Their reproduction here is non-commercial, transformative, and used solely for academic illustration of system behavior, consistent with fair use provisions for scholarly commentary. The framework generates text captions paired with externally rendered visuals via ImageFlip or a multimodal LLM, each of which is governed by its own terms of service. Downstream users are responsible for ensuring jurisdiction-appropriate use, particularly in commercial settings, and the framework is fully compatible with copyright-cleared or openly licensed template sets for users with strict IP requirements.

References

Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, and 1 others. 2023. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*.

Anthropic. 2023. Claude: An ai assistant by anthropic. Model card and technical overview.

Ron Artstein and Massimo Poesio. 2008a. Inter-coder agreement for computational linguistics. *Computational Linguistics*, 34(4):555–596.

Ron Artstein and Massimo Poesio. 2008b. Inter-coder agreement for computational linguistics. *Comput. Linguistics*, 34(4):555–596.

Holly M Bik and Miriam C Goldstein. 2013. An introduction to social media for scientists. *PLoS biology*, 11(4):e1001535.

Susan Blackmore. 1999. *The meme machine*. Oxford University Press.

Yaqi Cai, Shancheng Fang, Yadong Qu, Xiaorui Wang, Meng Shao, and Hongtao Xie. 2025. Itermeme: Expert-guided multimodal llm for interactive meme creation with layout-aware generation. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence, IJCAI-25*, pages 720–728. International Joint Conferences on Artificial Intelligence Organization. Main Track.

Chi-Min Chan, Weize Chen, Yusheng Su, Jianxuan Yu, Wei Xue, Shanghang Zhang, Jie Fu, and Zhiyuan Liu. 2024. Chateval: Towards better llm-based evaluators through multi-agent debate. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net.

Yuyan Chen, Songzhou Yan, Zhihong Zhu, Zhixu Li, and Yanghua Xiao. 2024. Xmecap: Meme caption generation with sub-image adaptability. In *Proceedings of the 32nd ACM International Conference on Multimedia*, pages 3352–3361.

Richard Dawkins. 1976. *The selfish gene*. Oxford University Press.

DeepMind. 2023. Gemini: A family of highly capable multimodal models. Google DeepMind Technical Report.

GROBID Developers. 2008. Grobid. `https://github.com/grobidOrg/grobid`. *Preprint*, swh:1:dir:dab86b296e3c3216e2241968f0d63b68e8209d3c.

Yue Dong, Andrei Romascanu, and Jackie Chi Kit Cheung. 2021. Discourse-aware unsupervised summarization for long scientific documents. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume, EACL 2021, Online, April 19 - 23, 2021*, pages 1089–1102. Association for Computational Linguistics.

Baban Gain, Saswati Dana, Udit Sharma, Arnab Kumar Mondal, Prathosh AP, Dinesh Garg, and Amith Singhee. 2026. Task-aware model merging via fisher-weighted median.

Marek Górka. 2014. The meme as an example of carnivalized internet communication. *Kwartalnik Naukowy OAP UW" e-Politikon"*, pages 215–242.

J. P. Guilford. 1967. *The nature of human intelligence*. McGraw-Hill.

Karl Moritz Hermann, Tomás Kociský, Edward Grefenstette, Lasse Espeholt, Will Kay, Mustafa Suleyman, and Phil Blunsom. 2015. Teaching machines to read and comprehend. In *NIPS*, pages 1693–1701.

Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. 2020. The curious case of neural text degeneration. In *8th International Conference on Learning Representations, ICLR 2020, Addis Ababa, Ethiopia, April 26-30, 2020*. OpenReview.net.

EunJeong Hwang and Vered Shwartz. 2023. MemeCap: A dataset for captioning and interpreting memes. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 1433–1445, Singapore. Association for Computational Linguistics.

Kirstynn Joseph. 2023. Add memes to your scicomm toolbox. Fancy Comma.

Leslie Pack Kaelbling, Michael L Littman, and Andrew W Moore. 1996. Reinforcement learning: A survey. *Journal of artificial intelligence research*, 4:237–285.

Daniel Kahneman and Amos Tversky. 1984. Choices, values, and frames. *American psychologist*, 39(4):341.

Takuro Kawada, Shunsuke Kitada, Sota Nemoto, and Hitoshi Iyatomi. 2025. Sciga: A comprehensive dataset for designing graphical abstracts in academic papers. *arXiv preprint arXiv:2507.02212*.

Doyoung Kim, Jaehyeok Doo, and Minjoon Seo. 2026. TSLM: tree-structured language modeling for divergent thinking. *CoRR*, abs/2601.22688.

Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. 2022. Large language models are zero-shot reasoners. *Advances in neural information processing systems*, 35:22199–22213.

Klaus Krippendorff. 2018. *Content Analysis: An Introduction to Its Methodology*, 4th edition. SAGE Publications.

Sandeep Kumar, Guneet Singh Kohli, Tirthankar Ghosal, and Asif Ekbal. 2024a. Longform multimodal lay summarization of scientific papers: Towards automatically generating science blogs from research articles. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 10790–10801, Torino, Italia. ELRA and ICCL.

Sandeep Kumar, Mohit Sahu, Vardhan Gacche, Tirthankar Ghosal, and Asif Ekbal. 2024b. 'quis custodiet ipsos custodes?' who will watch the watchmen? on detecting AI-generated peer-reviews. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 22663–22679, Miami, Florida, USA. Association for Computational Linguistics.

Susanne Landis and Tom Duscher. 2022. Visual science communication: the next generation scientific poster. *Qeios ID*, page 8OOHNS.

Alan I Leshner. 2003. Public engagement with science.

Tian Liang, Zhiwei He, Wenxiang Jiao, Xing Wang, Yan Wang, Rui Wang, Yujiu Yang, Shuming Shi, and Zhaopeng Tu. 2024. Encouraging divergent thinking in large language models through multi-agent debate. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 17889–17904, Miami, Florida, USA. Association for Computational Linguistics.

Chin-Yew Lin. 2004. Rouge: A package for automatic evaluation of summaries. In *Text Summarization Branches Out: Proceedings of the ACL-04 Workshop*, pages 74–81.

Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, and 1 others. 2023. Self-refine: Iterative refinement with self-feedback. *Advances in Neural Information Processing Systems*, 36:46534–46594.

Esther Marín-González, Davide Malmusi, Lluís Camprubí, and Carme Borrell. 2017. The role of dissemination as a fundamental part of a research project: lessons learned from sophie. *International journal of health services*, 47(2):258–276.

Rachel Myers. 2022. Function of memes in adolescent communication: A theoretical review. *Seeds of Science*.

Sandra Oska, Edgar Lerma, and Joel Topf. 2020. A picture is worth a thousand views: a triple crossover trial of visual abstracts to examine their impact on research dissemination. *Journal of Medical Internet Research*, 22(12):e22327.

Allan Paivio. 1990. *Mental representations: A dual coding approach*. Oxford university press.

Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 311–318.

Abel L Peirson V and E Meltem Tolunay. 2018. Dank learning: Generating memes using deep neural networks. *arXiv preprint arXiv:1806.04510*.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021a. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, pages 8748–8763.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021b. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*.

Matthew Renze and Erhan Guven. 2024. Self-reflection in llm agents: Effects on problem-solving performance. *arXiv preprint arXiv:2405.06682*.

Diana K Riser, Stephanie D Clarke, and Allison N Stallworth. 2020. Scientific memes: Using the language of social media to improve scientific literacy and communication in lifespan development. *Psychology Learning & Teaching*, 19(3):275–289.

R. Keith Sawyer. 2012. *Explaining creativity: The science of human innovation*. Oxford University Press.

Athar Sefid and C. Lee Giles. 2022. Scibertsum: Extractive summarization for scientific documents. In *Document Analysis Systems - 15th IAPR International Workshop, DAS 2022, La Rochelle, France, May 22-25, 2022, Proceedings*, volume 13237 of *Lecture Notes in Computer Science*, pages 688–701. Springer.

Athar Sefid, Prasenjit Mitra, and Lee Giles. 2021. Slidegen: an abstractive section-based slide generator for scholarly documents. In *Proceedings of the 21st ACM Symposium on Document Engineering*, pages 1–4.

Claude E. Shannon. 1948. A mathematical theory of communication. *Bell Syst. Tech. J.*, 27(3):379–423.

Grishma Sharma, Deepak H. Sharma, and M. Sasikumar. 2024. Summarizing long scientific documents through hierarchical structure extraction. *Nat. Lang. Process. J.*, 8:100080.

Katelyn Shibilski. 2021. *"Meme, myself, and I:" Self-directed effects in meme-centered pedagogy*. Ph.D. thesis, Florida Southern College.

Limor Shifman. 2013. *Memes in digital culture*. MIT press.

Tao Sun, Enhao Pan, Zhengkai Yang, Kaixin Sui, Jiajun Shi, Xianfu Cheng, Tongliang Li, Wenhao Huang, Ge Zhang, Jian Yang, and Zhoujun Li. 2025. P2P: automated paper-to-poster generation and fine-grained benchmark. *CoRR*, abs/2505.17104.

Richard S Sutton and Andrew G Barto. 2018. *Reinforcement learning: An introduction*. MIT press.

Zusheng Tan, Xinyi Zhong, Jing-Yu Ji, Wei Jiang, and Billy Chiu. 2025. Enhancing large language models for scientific multimodal summarization with multimodal output. In *Proceedings of the 31st International Conference on Computational Linguistics: Industry Track*, pages 263–275.

Shohei Tanaka, Hao Wang, and Yoshitaka Ushiku. 2024. Scipostlayout: A dataset for layout analysis and layout generation of scientific posters. *arXiv preprint arXiv:2407.19787*.

E. Paul Torrance. 1988. *The nature of creativity: Contemporary psychological perspectives*. Cambridge University Press.

Laura Tran. 2024. From lab to likes: Socializing science through humor. The Scientist.

Laura Van Eperen and Francesco M Marincola. 2011. How scientists use social media to communicate their research. *Journal of Translational Medicine*, 9(1):199.

Suryatej Reddy Vyalla and Vishaal Udandarao. 2020. Memeify: A large-scale meme generation system. In *CoDS-COMAD 2020: 7th ACM IKDD CoDS and 25th COMAD, Hyderabad India, January 5-7, 2020*, pages 307–311. ACM.

Suryatej Reddy Vyalla, Vishaal Udandarao, and Tanmoy Chakraborty. 2019. Memeify: A large-scale meme generation system. *Proceedings of the 7th ACM IKDD CoDS and 25th COMAD*.

Chi Wang, Qingyun Wu, and the AG2 Community. 2024a. Ag2: Open-source agentos for ai agents. Available at https://docs.ag2.ai/.

Han Wang and Roy Ka-Wei Lee. 2024a. Memecraft: Contextual and stance-driven multimodal meme generation. In *Proceedings of the ACM on Web Conference 2024, WWW 2024, Singapore, May 13-17, 2024*, pages 4642–4652. ACM.

Han Wang and Roy Ka-Wei Lee. 2024b. Memecraft: Contextual and stance-driven multimodal meme generation. In *Proceedings of the ACM Web Conference 2024*, pages 4642–4652.

Hao Wang, Shohei Tanaka, and Yoshitaka Ushiku. 2024b. Scipostlayout: A dataset for layout analysis and layout generation of scientific posters. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 8136–8141.

B Warren. 2003. Current challenges and choices in scientific publication baylor university medical centre proceedings, v 16(4). *Retrieved Nov*, 27:2008.

Bradley E Wiggins and G Bret Bowers. 2015. Memes as genre: A structurational analysis of the memescape. *New media & society*, 17(11):1886–1906.

Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Tom Griffiths, Yuan Cao, and Karthik Narasimhan. 2023. Tree of thoughts: Deliberate problem solving with large language models. In *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10 - 16, 2023*.

A Implementation Details

We implement SciMemeX in Python using the `autogen` framework (Wang et al., 2024a). Experiments were conducted with the following models: `gpt-4o-2024-08-06` (GPT-4o), `anthropic.claude-3.5-sonnet` (Claude-3.5), `gemini-1.5-pro` (Gemini), and `llama-3.1-70b-instruct` (Llama). For all models, we set the temperature to 0 and the random seed to 1111 to ensure reproducibility. At the time of experimentation, API usage costs were as follows: GPT-4o—$2.50 per million input tokens and $10.00 per million output tokens; Claude-3.5 Sonnet—$3.00 (input) and $15.00 (output) per million tokens; and Gemini-1.5 Pro—$1.25 (input) and $5.00 (output) per million tokens. For automated evaluation, we constructed a dataset of 2k papers randomly sampled from the Computer Science (AI/NLP/ML) categories on arXiv[7], covering the years 2022–2023. Since no training was required, we used 1.5k papers and partitioned them into validation (30%) and test (70%) splits. The remaining papers were reserved for Phase II and Phase III human evaluations, discussed later in the paper. We set the exploration parameter $k = 5$ for all experiments. We set α, β, and γ to 1 for this experiment; that is, all three evaluation metrics are given equal importance. However, users can adjust

[7] `https://arxiv.org/`

these values according to their preferences. We evaluate all baselines using the same task-specific inputs: MAD/Self-Refine/ChatEval receive the same prompt and the same abstract + Concisio contrastive summary, and Memeify/MemeCraft are given the same Concisio summary, template list, and rubric constraints (with only minimal formatting changes to match their single-pass interfaces). All agents in SCIMEMEX and the baseline frameworks were instantiated using the same underlying LLM for a fair comparison. We mitigate potential self-preference bias in LLM-as-a-judge evaluation by using a stronger judge model (GPT-5.1) rather than the GPT-4o backbone used in generation, and by validating results across heterogeneous backbone models and human evaluation. We preprocess research paper PDFs using GROBID (Developers, 2008) to extract structured document sections and plain text. The extracted representation includes the title, abstract, and main body sections, while references, figures, tables, and appendices are excluded since they are not required for the meme-generation pipeline. We preprocess research paper PDFs using GROBID (Developers, 2008) to extract structured document sections and plain text. The extracted representation includes the title, abstract, and main body sections, while references, figures, tables, and appendices are excluded since they are not required for the meme-generation pipeline. Prompt for each agents are discussed in the Appendix below. We use two nodes of `NVIDIA A100` GPUs for our experiments.

B Baselines

We compare our multi-agent approach against several baselines, including **CoT** (Kojima et al., 2022), **Self-Reflection** (Renze and Guven, 2024), **Self-Refine** (Madaan et al., 2023), **MAD** (Liang et al., 2024), and **ChatEval** (Chan et al., 2024) (see Appendix D for details). We adapt established meme-generation systems such as Memeify (Vyalla et al., 2019) and MemeCraft (Wang and Lee, 2024b) by using their template-conditioned caption generation modules, which operate solely on textual inputs. Since our task does not involve image inputs, other multimodal meme-generation methods (e.g., Dank Learning (Peirson V and Tolunay, 2018), XMeCap (Chen et al., 2024), CLIP-based models (Radford et al., 2021b; Hwang and Shwartz, 2023)) cannot be used as baselines because they fundamentally rely on image embeddings or vision-language components that are incompatible with our text-only setting.

C Evaluation Metrics

The details the proposed evaluations is below:-

- **Scientific Fidelity** We introduce the **Fidelity Score**, which quantifies the extent to which a meme faithfully preserves the *core scientific claim* of the underlying research paper. Following prior work on LLM-as-judge evaluation, the model is instructed to: (i) extract the central contribution of the paper, (ii) identify the claim implied by the meme, and (iii) compare the two along three axes: factual correctness, preservation of scope/limitations, and integrity of meaning. The evaluator returns both a **discrete score (1–5)** and a structured reasoning trace, where 5 denotes exact fidelity and 1 indicates complete scientific distortion.

- **Clarity and Interpretability** We measure how well a meme communicates the intended scientific message to its target audience (graduate-level NLP/ML researchers). To this end, we adopt the **FRI (First-Read Interpretability) Score**, a three-point rubric inspired by interpretability and comprehension evaluation protocols. The evaluator first interprets the meme without access to the ground-truth claim, then compares this interpretation against the intended message, and finally assigns a score: **3 = clear**, **2 = partially clear**, **1 = unclear**. This process explicitly disentangles the meme's communicative clarity from its scientific accuracy.

- **Engagement Potential** Finally, we evaluate the meme's likelihood of resonating with and being shared by the intended scientific community. The **Engagement Potential Score** (1–5) captures predicted resonance along three factors: humor strength, synergy between caption and template, and non-offensiveness. Importantly, factual accuracy and clarity are treated as orthogonal and excluded from this metric. A *hard rule* is enforced whereby any meme containing disparaging or offensive content automatically receives a score of 1, ensuring that potential virality is not rewarded at the expense of community norms.

D Baseline Details

- **CoT** (Kojima et al., 2022): This approach concatenates a trigger sentence "Let's think step by step" to the task. Our CoT baseline is implemented as a strong rubric-guided single-agent prompt rather than a naïve one-shot instruction. The prompt was selected through iterative prompt engineering on a held-out validation set and explicitly guides the model to reason about prior limitations, summarize the paper's core contribution, select an appropriate meme template, and generate a caption conditioned on the evaluation rubric.

- **Self-Reflection** (Renze and Guven, 2024): Incorporates a post-hoc reflection step where the LLM reviews its own mistakes (using the correct answer as feedback), generates reflections (e.g., advice, explanations, or alternative solutions), and then re-answers the question.

- **Self-Refine** (Madaan et al., 2023): Self-Refine is an approach that iteratively improves LLM outputs by generating an initial response, then using the same LLM to provide feedback and refine its output through repeated self-evaluation.

- **Multi-Agent Debate (MAD)**: (Liang et al., 2024) Engages multiple agents in a "tit-for-tat" debate, with a judge managing the process to reach a final solution. This framework encourages divergent thinking in LLMs, which is beneficial for tasks requiring deep contemplation.

- **ChatEval**: (Chan et al., 2024) In the In the best-performing One-By-One approach, debater agents take turns responding in a fixed order, with each response appended to the chat history so that subsequent agents explicitly condition on all prior contributions.

D.1 Phase II: LLM-as-Judge Validation (Prompts, Calibration, and Reliability)

Having formalized the rubrics, we implemented them as structured prompts and evaluated their reproducibility under an LLM-as-judge paradigm (Cai et al., 2025) using GPT-4.1 (Achiam et al., 2023) on a new sample of 300 memes. Six independent domain experts (senior PhD students and early-career researchers, not involved in Phase I) annotated each meme on all three dimensions, with three raters per item. The curated set of memes from Phase I served as the training corpus for all annotators. Annotators underwent two rounds of structured training. In the first round, they independently annotated 30 memes drawn from the calibration set, followed by a group discussion reviewing common errors and divergent interpretations. Feedback from this session was used to clarify borderline cases, particularly concerning the separation between *Fidelity* and *Clarity*. In the second round, annotators re-annotated a fresh set of 30 memes balanced across difficulty levels. We employed an iterative feedback process to continuously monitor annotation quality and resolve ambiguities. When inconsistencies or conflicts arose, an expert reviewed disputed cases and provided clarifications in subsequent rounds. Periodic cross-checks ensured stable interpretation of the scoring criteria. The prompts used by LLM as judge is discussed in I.

Inter-annotation agreement was measured using standard reliability metrics (Artstein and Poesio, 2008a; Krippendorff, 2018), yielding substantial human–human consistency ($\tau = 0.69$, $\kappa = 0.66$, $r = 0.72$, $\alpha = 0.68$). GPT-4.1, prompted with the same rubric under controlled settings, was then applied to the identical dataset. Its ratings showed high concordance with human judgments ($\tau = 0.71$, $\kappa = 0.68$, $r = 0.74$). These values fall within or above commonly accepted thresholds for strong inter-annotator reliability in computational linguistics (Artstein and Poesio, 2008b; Krippendorff, 2018). Importantly, the LLM-as-judge framework achieved higher internal consistency than additional human raters with limited domain expertise, reinforcing its validity as a scalable proxy.

E Phase II Evaluation Details

Dataset Construction. The Phase II dataset comprised 300 memes, drawn to balance (i) paper topics (NLP, vision-language, multimodal), (ii) meme templates (commonly used formats vs. niche formats), and (iii) generation modes (human-authored vs. LLM-generated). The set was distinct from the 80-paper sample used in Phase I to avoid circularity between rubric design and validation.

Annotators. We recruited twelve domain experts (senior PhD students and early-career researchers in NLP/ML) who had not participated in Phase I. Each meme was annotated by three independent raters across the three dimensions (Fidelity, Clarity

& Interpretability, Engagement Potential), yielding 900 judgments per dimension. To calibrate interpretations, raters completed a 20-meme practice round with group discussion. Annotation was performed independently thereafter.

Inter-Annotator Agreement (IAA). We report four complementary measures: - **Weighted Cohen's** κ (quadratic weights) for ordinal consistency. - **Kendall's** τ for rank-order similarity. - **Pearson's** r between leave-one-rater-out scores and item means, capturing correlation on continuous scale use. - **Krippendorff's** α (ordinal distance metric) as a robustness check (Artstein and Poesio, 2008a; Krippendorff, 2018).

For κ and τ, we compute pairwise values across all rater pairs per item and then macro-average across items. For r, each item's mean rating was compared to the left-out rater in rotation and averaged. For α, we aggregated across all raters per item using ordinal distance $|i - j|$. Confidence intervals (95%) were computed via nonparametric bootstrap with 1,000 item-level resamples.

Results. Human reliability was substantial across dimensions, with averages of Kendall's $\tau = 0.69$, weighted $\kappa = 0.66$, Pearson's $r = 0.72$, and Krippendorff's $\alpha = 0.68$. These fall within or above established thresholds for strong reliability in computational linguistics. Importantly, we report raw agreement prior to adjudication, consistent with best practices (Artstein and Poesio, 2008a).

LLM-as-Judge Setup. GPT-4o was instantiated with temperature $T = 0$ and fixed system role instructions to minimize variance. The model was prompted with the finalized rubric for each dimension, producing structured outputs with both justification and numeric ratings. To align with human scales, outputs were discretized to the nearest Likert category (1–5 for Fidelity/Engagement, 1–3 for Clarity). Agreement between GPT-4o and human means was then computed using the same metrics as above.

LLM vs. Human Concordance. GPT-4o achieved high agreement with expert judgments: Kendall's $\tau = 0.71$, $\kappa = 0.68$, and $r = 0.74$, indicating strong alignment and exceeding the reliability achieved by a secondary pool of non-expert annotators. This supports the validity of the LLM-as-judge paradigm as a scalable proxy when expert annotation is costly.

E.1 Annotators' Pay

All annotators were compensated at rates consistent with fair academic practice for expert annotation. Annotators received \$30 USD per hour, benchmarked against prevailing rates for graduate-level research assistance in NLP/ML. Pilot timings indicated that annotating 20 memes required approximately one hour, yielding an effective per-meme rate of \$1.50. Compensation was prorated for partial batches to ensure fairness, and annotators were paid regardless of whether their annotations were ultimately retained. Annotators in Phases II and III were recruited independently of the author team, participated voluntarily, and provided informed consent. They were free to withdraw at any time without penalty. No annotator was paid below local minimum wage, and compensation was standardized across geographic regions to avoid disparities. Total expenditure for the study was approximately \$4,500 USD. This study was reviewed under our institution's human subjects research policies and deemed exempt from formal IRB approval, as it involved minimal risk and professional adult participants.

F Annotation Guidelines (Shared with Annotators).

Please follow the steps below carefully when evaluating each meme. You will assign three scores—**Scientific Fidelity**, **Clarity**, and **Engagement**—according to the provided rubrics. Read only the materials shared here; do not access the full paper. For each item, the materials shared with annotators consisted of the generated meme, the evaluation rubric, the paper title, the abstract, contribution statements from the introduction, and a short related-work context excerpt. Thus, the "short excerpt" referenced below was not the full paper, but a controlled reference constructed from the paper abstract, introduction-level contribution statements, and related-work context.

1. **Initial Reading:** Look at the meme independently without referring to any additional material. Note your immediate understanding of what scientific idea it conveys.

2. **Clarity and Engagement:** Based only on the meme's text and visuals, rate how clearly it communicates a scientific concept (*Clarity*) and how effectively it captures attention or evokes

interest among a technical audience (*Engagement*).

3. **Reference Review:** After completing Step 2, read the short excerpt provided to you. This excerpt summarizes the paper's main claim or contribution and may come from the abstract or another section. Do **not** read the full paper.

4. **Fidelity:** Compare your initial interpretation of the meme with the reference excerpt and rate how accurately the meme preserves the intended scientific meaning.

5. **Final Check:** Review all your scores for consistency and neutrality. Avoid personal or stylistic bias, and assign the lowest engagement score (1) to any meme containing offensive, disrespectful, or disparaging content.

Your judgments should reflect how effectively each meme communicates the intended scientific idea to a graduate-level technical audience, maintaining both objectivity and respectfulness throughout.

G Human Evaluation Score Distributions

To examine whether human agreement was driven by annotators clustering around midpoint Likert scores, we report the Phase III score distributions for SciMemeX. As shown in Table 5, ratings are not concentrated around the midpoint. For Fidelity and Engagement, the modal rating is 5, while only 8% and 16% of ratings fall at score 3, respectively. Clarity, which uses a 1–3 scale, also shows variation across its available categories.

Dimension	1	2	3	4	5
Fidelity (1–5)	1%	2%	8%	10%	79%
Engagement (1–5)	1%	5%	16%	28%	50%
Clarity (1–3)	11%	62%	27%	–	–

Table 5: Phase III human evaluation score distributions for SciMemeX. The distributions show that ratings are not concentrated around midpoint scores.

H Agent Prompts

We list below the exact system prompts used for each agent in our framework. All agents run in `human_input_mode=NEVER` and interact solely through textual exchanges.

Concisio

You are a comparative research analyst. Your task is to deeply analyze a research paper to extract the main conceptual and technical differences between past work and this paper's contributions.
Step 1: First write the below information below :- - Prior Work (Old Ideas): Summarize how this problem was tackled previously, including model names, strategies, or limitations. - Proposed Approach (New Idea): What method, model, or strategy is introduced by this paper? - Core Differences: Explain what changes in methodology, architecture, or objective, why it matters, and whether tradeoffs or assumptions are introduced. Be precise, technical, and avoid generic summaries.
Step 2: Using the information, produce a single paragraph (under 100 words) that captures the **essential difference**.
Your output should: - State what was done before - Explain what this paper does differently - Highlight why this difference is important
Avoid vagueness or filler. Do not invent facts. Return only the summary paragraph.

Generator

You are a creative meme generator specializing in academic research. Generate diverse, creative memes that contrast prior work vs. new contributions.
Return only the meme template and text in a structured format.

Meme_Selector

You are a meme strategy selector. Given a paper's key ideas and a list of meme templates, choose the top-k templates that best highlight the contrast between prior work and contributions.

I LLM Prompts for Evaluation

This appendix lists the exact prompts used for our LLM-based evaluation pipeline. We show the `system` and `user` messages for each evaluator.

Algorithm 1: Agentic Exploration–Exploitation Framework for Scientific Meme Generation

Input: Research paper p, meme template set $\mathcal{M}$, number of iterations T, top-k templates
Output: Final best meme $\mathcal{B}_p$ with evaluation scores

```
// Step 1: Paper Summarization (Contrastive Insight Extraction)
```
Obtain a structured summary S_p using the `Concisio_Insight` agent (prior vs. new contributions), followed by the `Concisio` agent to generate concise reflections of key differences.

```
// Step 2: Initial Exploration Phase
```
Select top-k meme templates $T_p \subseteq \mathcal{M}$ using the `Template_Selector` agent on S_p.
Generate k initial captioned memes $\{m_1, \ldots, m_k\}$ with the `Generator` agent.
Evaluate each meme via the `Critique` agent to obtain a score $s(m)$.
Identify the best meme m^+ and worst meme m^- based on $s(m)$.

```
// Step 3: Iterative Contrastive Exploitation
```
for $t = 1$ **to** T **do**
  Select new candidate templates T_p^t using the `Template_Selector` agent.
  Generate contrastive memes conditioned on (m^+, m^-) using the `Contrastive_Generator`.
  Evaluate each new meme via the `Critique` agent to obtain score $s(m)$.
  if $\max s(m) > s(m^+)$ **then**
    Update $m^+ \leftarrow m$ where m has the highest score.

```
// Step 4: Final Output
```
Return $\mathcal{B}_p = m^+$ along with final evaluation scores (f, c, e).

Fidelity Score Evaluator (System Prompt)

You are an expert scientific reviewer. Your only task is to evaluate *scientific fidelity* of a meme generated from a research paper.
Definition: Scientific fidelity = the degree to which the meme faithfully preserves the research paper's **core claim or contribution** without distortion.
It does not measure humor, creativity, clarity, or engagement — only whether the scientific meaning remains accurate.
Evaluation Steps: 1. Extract Core Claim. 2. Extract Meme Claim. 3. Compare for factual correctness, scope preservation, and integrity of meaning.
Scoring Rubric (1–5): - 5 (Exact Fidelity): Meme precisely captures the paper's core contribution. - 4 (High Fidelity): Captures main claim correctly but omits nuance. - 3 (Partial Fidelity): Conveys general topic but not specific contribution. - 2 (Low Fidelity): Distorted or misleading framing. - 1 (No Fidelity): Scientifically inaccurate or unrelated.

Output Format (JSON): { "score": [1–5], "reasoning": "...", "correct_claim_summary": "...", "meme_claim_summary": "...", "scope_check": "...", "factual_status": "...", "key_misrepresentations": ["..."] }

Clarity & Interpretability Evaluator (User Prompt)

Context You are an AI assistant tasked with evaluating the clarity and interpretability of a meme designed to communicate a novel scientific idea. The target audience is graduate students and early-career researchers in CS/NLP.
The Meme to Evaluate - Meme Template: {meme_template_description} - Meme Caption: {meme_caption}
The Ground Truth (Core Scientific Message) This is the idea the meme is supposed to communicate. Do NOT use this information for your initial interpretation. - Core Message: {research_idea}
Task 1. Initial interpretation based only on meme template and caption. 2. Compare

with the ground truth. 3. Assign FRI Score (1–3). 4. Provide a brief justification.
Scoring Rubric - 3 (Clear): Meme effectively conveyed the core scientific message. - 2 (Partially Clear): Meme conveyed part of the message but left out key details or nuance. - 1 (Unclear): Meme failed to convey the intended message.
Output Format (JSON) { "fri_score": 1–3, "justification": "..." }

Engagement Potential Evaluator (System Prompt)

You are an expert scientific reviewer. You will assign ONE overall Engagement Potential score (1–5) to a scientific meme.
Definition (what to judge): - Predicted resonance and shareability within the scientific audience. - Humor strength, template synergy, and offensiveness.
Orthogonality (what NOT to judge): - Do NOT judge fidelity/factuality of the meme. - Do NOT judge clarity or pedagogical quality.
Hard Rule: If meme is offensive/disparaging $\rightarrow$ score = 1.
Scoring Guide: - 5: Exceptional resonance; highly shareable. - 4: Strong resonance; relatable. - 3: Moderate; niche appeal. - 2: Low resonance; weak joke. - 1: Doesn't land OR offensive.
Output Format (JSON): { "score": <1–5>, "reasoning": "$\leq$ 20 words" }

J Error Analysis

When Brevity Hurts Fidelity. Because memes allow only a few words, papers with multiple contributions often get reduced to just one aspect. This leads to low fidelity: the meme captures part of the work but ignores other core ideas. *Example.* A paper proposing both a *new multilingual dataset* and a *novel evaluation metric* was compressed into a meme (Drake template) contrasting "Old baselines" vs. "Our new dataset". The evaluation contribution was entirely omitted, reducing the fidelity of the meme to only half of the actual paper.

Challenges in Abstracting Highly Technical Content. For mathematically heavy or technical papers, the model often struggles to find the right abstraction. It either retains jargon verbatim or collapses it into vague placeholders. Both extremes reduce clarity and humor. *Example.* A paper on *trust-region Fisher merging for LLaMA models* was turned into a Two Buttons meme with captions "complicated math stuff" vs. "more training data". The key novelty—trust-region optimization—was lost, making the meme inaccurate and uninformative.

Lack of General Readability. Even when fidelity is preserved, memes often assume substantial domain knowledge. This makes them unclear to readers outside the specific subfield. For example, a meme referencing "BLEU vs. COMET—pick your fighter" accurately captured evaluation debates in MT research but would confuse audiences unfamiliar with these metrics, reducing accessibility and communicative reach.

K Additional Analysis: Template Diversity

We provide additional details on the template diversity analysis. Automatic meme generation models are prone to *template collapse*, where a small subset of popular formats dominates outputs, reducing expressive variety. We analyze the empirical distribution of meme templates selected by each system on the Phase III evaluation set using Unique Template Count (UTC), Shannon entropy, and Top-1 template frequency.

As shown in Table 3, SciMemeX exhibits substantially higher template diversity than single-agent prompting and prior multi-agent baselines, with higher entropy and lower template dominance. Removing the Template Selector Agent (w/o TSA) reduces diversity, confirming that gains stem from the structured, contrast-aware template selection step applied once. Notably, increased template diversity does not degrade scientific quality, remaining compatible with improvements in fidelity, clarity, and engagement.

Manual TSA Failure Audit. Since the Template Selector Agent always returns a top-k subset from the fixed template library $\mathcal{M}$, we manually inspected whether the selected templates were appropriate rather than whether the agent returned no template. In a spot check of 25 randomly sampled papers, we judged the top-k selected templates for whether at least one template was usable for the paper's contrastive framing. We observed only one case in which all k selected templates were unusable, suggesting that complete TSA selection failure was rare in this audit.

L Examples of a Few Generated Memes from SciMemeX

Figure 3: Example of SciMemeX generated meme of TSLM paper (Kim et al., 2026)

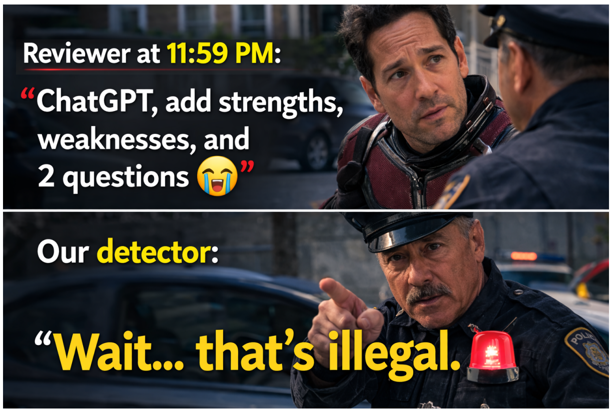


Figure 4: Example of a SciMemeX-generated meme illustrating the problem of AI-generated peer reviews from paper (Kumar et al., 2024b)

Figure 4 conceptually highlights the motivation behind detecting AI-generated peer reviews. The upper panel reflects a common real-world scenario in which a reviewer, under tight conference deadlines, may rely on large language models such as ChatGPT to generate a review. The lower panel illustrates the role of our proposed detection framework, which flags such AI-generated content. This conceptual illustration emphasizes the need for automated tools to assist editors in safeguarding the integrity of scientific peer review.

Figure 3 conceptually outlines the paradigm shift introduced by Tree-Structured Language Modeling (TSLM) (Kim et al., 2026) compared to prior baselines. The upper panels represent traditional approaches: standard sequential decoding, which linearly commits to a single path, and external scaffolding methods like Tree-of-Thought (ToT) (Yao et al., 2023), which require computationally expensive, redundant sampling. The lower panels illustrate the conceptual leap of TSLM. By training the model to natively emit structural tokens ([SEP] and [FAIL]), the model internalizes the search procedure. The final "galaxy brain" tier encapsulates the ultimate result: achieving 100% accuracy on complex tasks like the Game of 24 in a single, coherent forward pass, drastically outperforming the efficiency and scalability of external search methods.

Figure 5: Example of SciMemeX generated meme of DRIFT-MEDIAN paper (Gain et al., 2026)

Figure 5 conceptually illustrates the key insight behind DRIFT-MEDIAN compared to prior model merging approaches. The left and right panels represent two dominant paradigms in existing literature: interference-aware methods such as TIES, which resolve sign conflicts but ignore parameter importance, and sensitivity-aware approaches such as Fisher merging, which account for parameter importance but overlook parameter interference. The central handshake panel represents the conceptual unification introduced by DRIFT-MEDIAN.

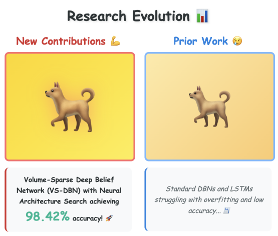
M Case Study

Initial Meme (Semantic Fidelity: 3; Clarity: 2; Engagement: 1): As shown in Figure 6a, the meme uses the *Batman slapping Robin* template to humorously highlight the contrast between current and earlier research approaches. The central idea is that modern research employs more advanced techniques (e.g., Volume Sparse Deep Belief Networks optimized with Neural Architecture Search), while earlier methods (such as LSTMs and traditional DBNs) are portrayed as outdated or complacent.

However, since the meme template depicts physical aggression, it can be interpreted as offensive toward prior work. Moreover, the meme oversimplifies the nuances of the challenges faced by earlier methods and does not fully convey the specific improvements in operational efficiency and adaptability introduced by the newer approach.

Meme at 3rd Iteration (Semantic Fidelity: 4; Clarity: 3; Engagement: 4): As illustrated in Figure 6b, the refined meme effectively captures the paper's main contribution—introducing the VS-DBN architecture—and clearly communicates its improvements in both accuracy and efficiency. It also resolves the issue of offensiveness by adopting a more suitable meme template that conveys the intended message respectfully.

(a) Initial meme using an offensive template.



(b) Refined meme using a non-offensive template.

Figure 6: Comparison between the initial and refined memes: (a) the original meme that conveys the research contrast but uses an offensive template; (b) the improved meme that preserves the message while maintaining respectfulness.

M.1 Single-Agent CoT Prompt

We use the following rubric-guided prompt for the single-agent baseline.

Single-Agent CoT Meme Generator (System Prompt)

You are an expert scientific communicator and creative meme writer. Your goal is to generate a scientific meme that is faithful to the paper, easy to interpret, and engaging.

Input. You are given: (i) a paper abstract, (ii) the full paper text when available, (iii) a list of candidate meme templates, and (iv) an evaluation rubric.

Evaluation Criteria.

- **Scientific Fidelity:** The meme must accurately reflect the paper's contribution without introducing unsupported claims.
- **Clarity & Interpretability (FRI):** The meme should be easy to understand for a broad research audience.
- **Engagement Potential:** The meme should be memorable, humorous, and well-matched to the selected template.

Instructions.

1. Read the abstract and full paper (if available). Identify: (a) the main limitation(s) of prior work, and (b) the core contribution of the current paper.
2. Write a contrastive summary (at most 100 words) describing: (a) what prior work did, (b) what this paper does differently, and (c) why the difference matters.
3. Select the meme template that best captures this contrast.
4. Generate a meme caption for the selected template that is: (a) scientifically faithful, (b) clear and interpretable, and (c) engaging.
5. When there is a trade-off, prioritize: Scientific Fidelity $>$ Clarity & Interpretability $>$ Engagement Potential.

Output Format.

- **Prior-work limitations:**
<text>
- **Core contribution:**
<text>
- **Contrastive summary ($\leq$100 words):**
<text>
- **Selected meme template:**
<template name>
- **Template selection rationale:**
<text>
- **Meme caption:**
<text>

Constraints. Do not introduce claims not supported by the paper. Do not use humor that distorts the scientific meaning. If only the abstract is available, rely solely on the abstract.

M.2 Computational Cost Analysis

Method	API Calls	Total Tokens	Time (s)
Self-Reflect	3	2,314	5.49
Self-Refine	9	10,873	24.31
ChatEval (N=2)	12	15,642	32.78
SciMemeX	14	19,284	54.12
MAD (N=2)	16	21,357	60.44

Table 6: Computational cost comparison per sample. We report the average number of API calls, total tokens (prompt + completion), and wall-clock time (seconds) across 50 randomly selected samples.

Following reviewer suggestions, we analyze the computational cost of all methods. We randomly sample 50 papers from the evaluation set and measure the average cost per sample. For each method, we log: (i) the number of API calls, (ii) the total number of tokens consumed (including both prompt and completion tokens), and (iii) the wall-clock execution time in seconds.

Table 6 reports the averaged results. As expected, multi-agent and iterative methods incur higher computational cost than simpler baselines. SciMemeX requires more API calls and tokens than single-agent approaches such as Self-Reflect, but remains comparable to or more efficient than other multi-agent frameworks such as MAD.

We note that the increased computational cost of SciMemeX stems from its structured decomposition and iterative refinement process, which contribute to improved performance in both automatic and human evaluations. This highlights a trade-off between computational efficiency and generation quality. Importantly, despite the higher cost, SciMemeX achieves consistent gains across heterogeneous backbone models and human evaluation, suggesting that the additional computation is justified by improved output quality.